**National University of Computer & Emerging Sciences**
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# OBJECT-ORIENTED PROGRAMMING LANGUAGE

## Project Report
## LUDO

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## Abstract

This project presents the development of a Lite Version of the classic Ludo game using C++. The main purpose of this project was to apply object-oriented programming concepts and basic game development logic in a simple console-based application.

The game includes core Ludo functionalities such as player turns, dice rolling, token movement, and basic winning conditions. Through this project, we focused on implementing user interaction, game rules, and logical flow in an organized manner.

The project helped improve programming skills, problem-solving abilities, and understanding of C++ concepts like classes, functions, loops, and conditional statements.

This Lite Ludo game provides an engaging way to combine learning with entertainment while demonstrating the practical use of programming fundamentals.


## Introduction

Ludo is a popular strategy-based board game enjoyed by people of all ages. The objective of the game is to move tokens from the starting point to the destination based on dice rolls while competing against other players. In this project, we developed a simplified or Lite version of Ludo in C++ to simulate the basic mechanics of the game in a console environment.
The project was designed to strengthen our understanding of programming by transforming a real-world board game into a software application.

Instead of implementing all advanced features of the original Ludo game, this version focuses on essential gameplay elements such as dice generation, player movement, turn handling, and result declaration. By using C++, we applied programming concepts including object-oriented design, modular programming, and logical decision-making.

This project serves as both an educational exercise and an enjoyable game simulation.

## Objectives

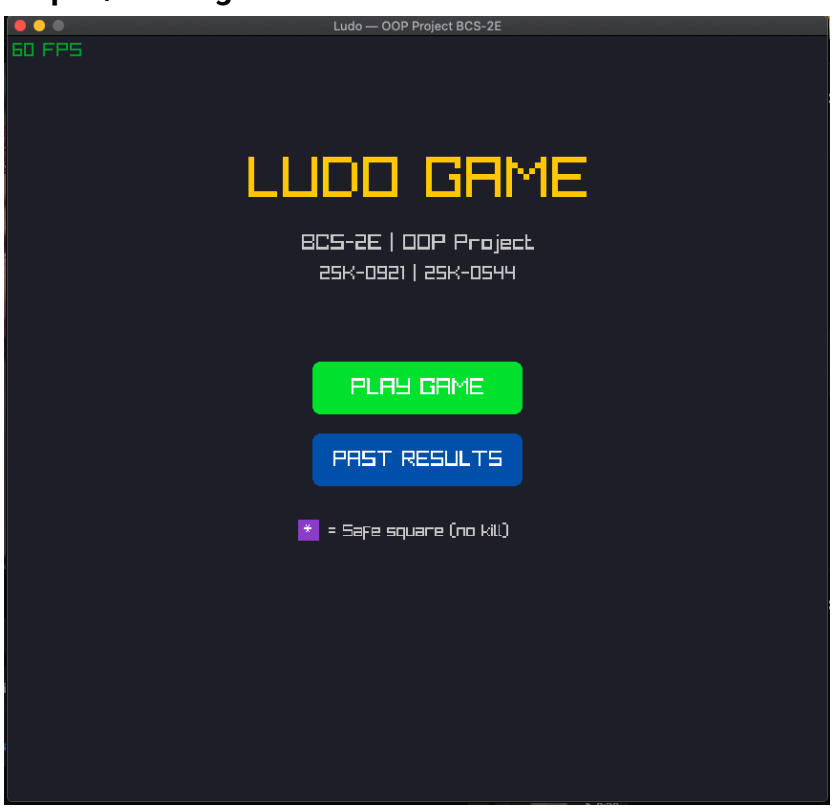
- To design and implement a simplified Ludo game using C++.
- To apply object-oriented programming concepts such as classes and functions.
- To develop logical game mechanics including dice rolling and player movement.
- To improve problem-solving and coding skills through practical implementation.

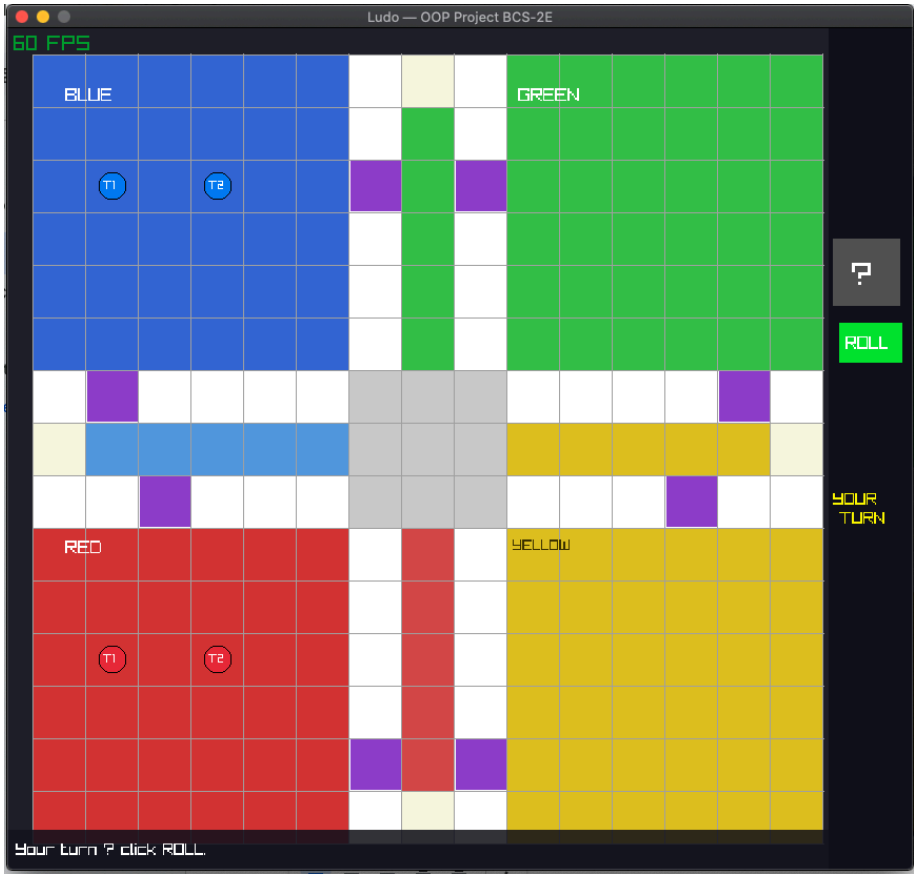
- To create an interactive console-based game for user engagement.
- To understand how real-world games can be converted into software applications.

## Implementation

- **Language:** C++
- **Compiler/IDE:** Code: Visual Studio Code, Dev C++
- **Key Features:**
    - **Dice Rolling System:** Random number generation to simulate dice rolls for each turn.
    - **Player Turn Management:** Organized sequence of turns between players.
    - **Token Movement:** Basic movement of player pieces according to dice values.
    - **Winning Condition:** Detection of the winning player based on reaching the destination first.
    - **Simplified Game Rules:** Focused on essential Ludo mechanics without advanced complexities.
    - **Modular Code Structure:** Functions and classes were used to keep the program organized and maintainable.

**Output / Testing and Results**






On the right side we have added roll button and we can see our number above the button or at the bottom left.

Here we have also dealt with file handling from the main menu if we want to see the past results we can just click on the past result buttons and the results are being read from the .txt file here.

```
ludo-yesy % ls
app
freesound_community-070766_2d6-rolling-dice-35977.mp3
ludo_results.txt
main.cpp
```

main.cpp | ludo_results.txt

ludo_results.txt

```
1   [2026-04-25 12:46:10] Winner: Player (Red)
2
```

## Conclusion

The Lite Ludo game project successfully demonstrated how a traditional board game can be transformed into a simple software application using C++. Through this project, we applied fundamental programming concepts such as object-oriented programming, logical decision-making, loops, and functions to create an interactive console-based game. The project enhanced our understanding of game development structure, user interaction, and problem-solving techniques. Although simplified, the game effectively achieved its purpose of combining learning with entertainment while strengthening practical coding skills.

## Limitations

- Advanced Ludo features such as multiple tokens, safe zones, and complex game rules were simplified or excluded.
- The game may support fewer customization options for players.
- Random dice generation may not perfectly replicate all real-world gameplay variations.
- Error handling and input validation may be basic in the Lite version.
- And most importantly if one user has not killed any token, and it cannot reach its home so the game at this level stucks over there and we have to restart the game.

## References


- Online C++ documentation and tutorials
- Basic Ludo game rules for gameplay logic understanding
- AI tools such as CLAUDE, GPT, GEMINI
- Youtube Channel: Programming Bugz